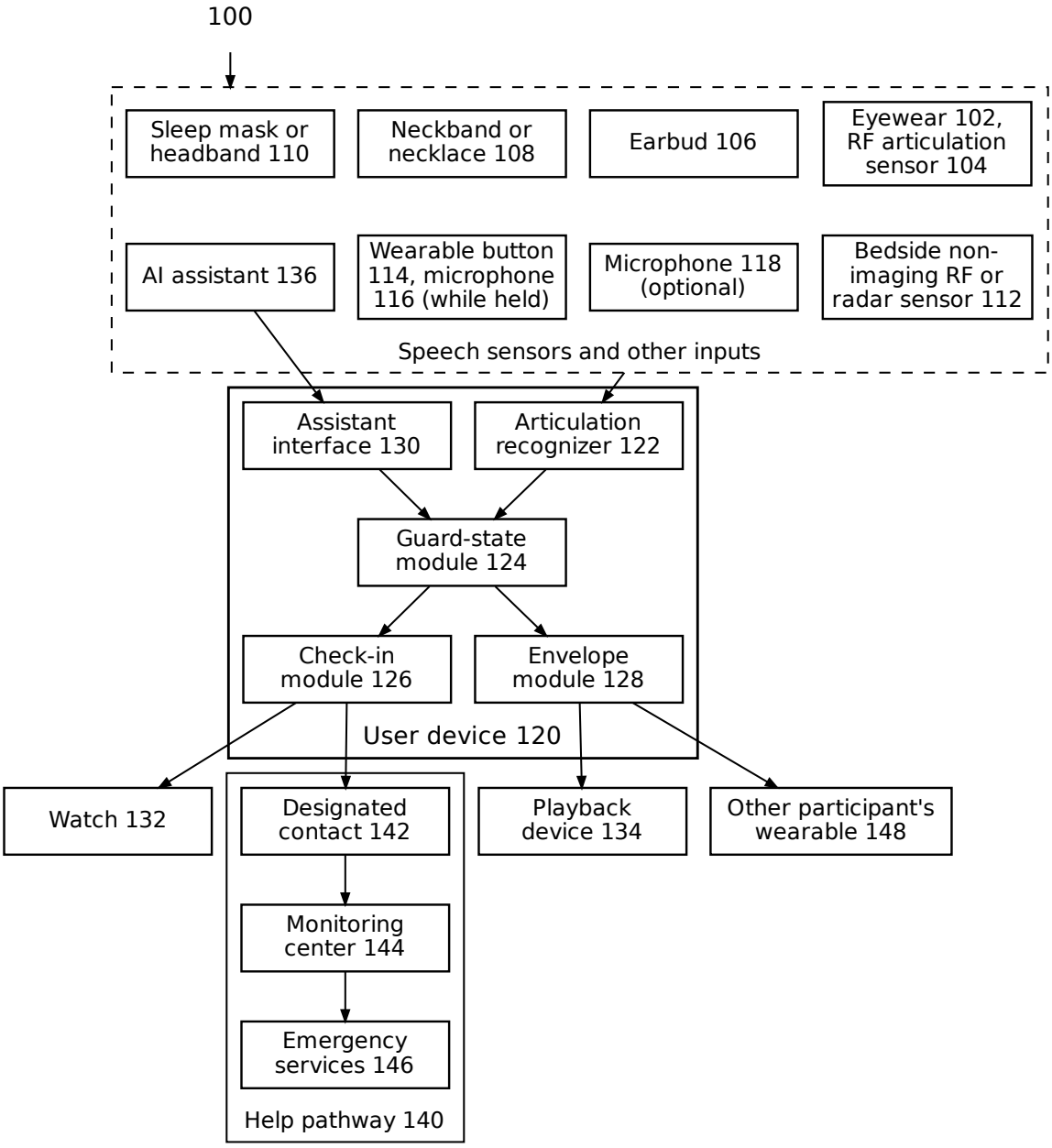
100
Sleep mask or headband 110
Neckband or necklace 108
Earbud 106
Eyewear 102, RF articulation sensor 104
AI assistant 136
Wearable button 114, microphone 116 (while held)
Microphone 118 (optional)
Bedside non-imaging RF or radar sensor 112
Speech sensors and other inputs
Assistant interface 130
Articulation recognizer 122
Guard-state module 124
Check-in module 126
Envelope module 128
User device 120
Watch 132
Designated contact 142
Playback device 134
Other participant's wearable 148
Monitoring center 144
Emergency services 146
Help pathway 140

FIG. 1

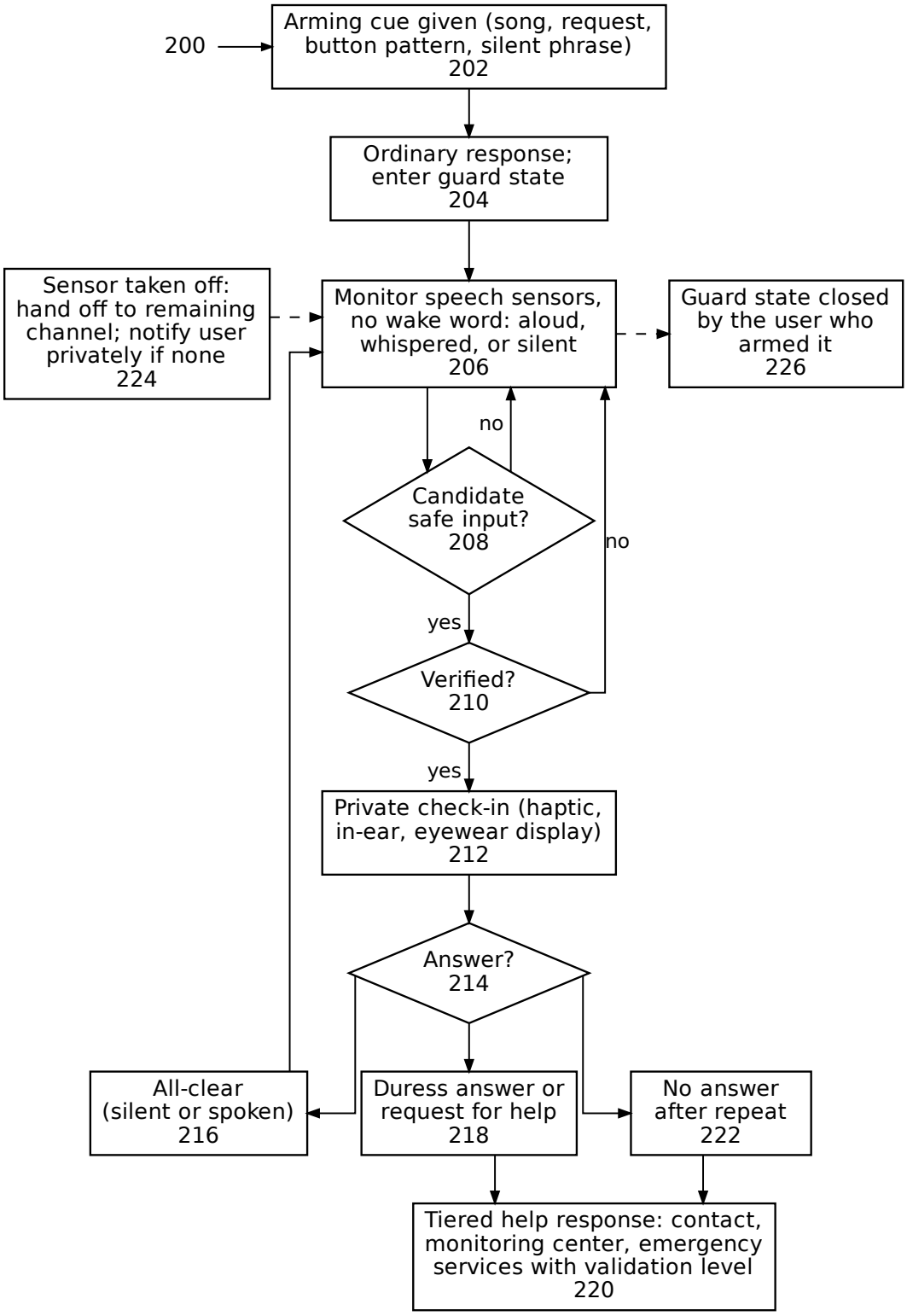
200
Arming cue given (song, request, button pattern, silent phrase)
202
Ordinary response; enter guard state
204
Sensor taken off: hand off to remaining channel; notify user privately if none
224
Monitor speech sensors, no wake word: aloud, whispered, or silent
206
Guard state closed by the user who armed it
226
no
Candidate safe input?
208
no
yes
Verified?
210
yes
Private check-in (haptic, in-ear, eyewear display)
212
Answer?
214
All-clear (silent or spoken)
216
Duress answer or request for help
218
No answer after repeat
222
Tiered help response: contact, monitoring center, emergency services with validation level
220

FIG. 2